



Beginner Level (Foundation)

Goal: Understand the basics of the C language, including syntax, data types, control structures, and functions.

1. Introduction to C Language

- **Topics:**
 - Overview of C language
 - Writing a simple C program
 - Understanding syntax and structure
- **YouTube Tutorial:**
 - [Introduction to C Programming by Geeky Shows](#)
- **Reference:**
 - C Programming Language – Tutorialspoint

2. Variables, Data Types, and Operators

- **Topics:**
 - Types of data types (int, float, char, etc.)
 - Operators (arithmetic, relational, logical)
 - Variables and constants
- **YouTube Tutorial:**
 - [Variables and Data Types in C by MyCodeSchool](#)
- **Reference:**
 - Variables and Data Types – GeeksforGeeks

3. Control Flow (if, else, loops)

- **Topics:**
 - if, else, switch statements
 - while, for, do-while loops
- **YouTube Tutorial:**
 - [Control Flow in C by LearnCodeOnline](#)
- **Reference:**
 - Control Flow - W3Schools



Intermediate Level (Enhance Your Skills)

Goal: Strengthen your understanding of functions, arrays, pointers, and memory management in C.

4. Functions and Recursion

- **Topics:**
 - Function declaration, definition, and calling
 - Recursion in C
- **YouTube Tutorial:**
 - [Functions in C by GeeksforGeeks](#)
- **Reference:**
 - Functions in C - GeeksforGeeks

5. Arrays and Strings

- **Topics:**
 - Single and multi-dimensional arrays
 - String manipulation
- **YouTube Tutorial:**
 - [Arrays in C by CodeWithHarry](#)
- **Reference:**
 - Arrays in C - Tutorialspoint

6. Pointers and Memory Management

- **Topics:**
 - Understanding pointers
 - Memory allocation using malloc() and free()
- **YouTube Tutorial:**
 - [Pointers in C by The Coding Train](#)
- **Reference:**
 - Pointers in C - GeeksforGeeks

Advanced Level (Become a Pro)

Goal: Master advanced concepts like structures, file handling, dynamic memory management, and system programming.

7. Structures and Unions

- **Topics:**
 - Defining and using structures
 - Unions and their usage
- **YouTube Tutorial:**
 - [Structures in C by CodeWithHarry](#)
- **Reference:**

- Structures in C - GeeksforGeeks

8. File Handling in C

- **Topics:**
 - File operations (reading, writing, and appending)
 - File pointers and file modes
- **YouTube Tutorial:**
 - [File Handling in C by The Learning Point](#)
- **Reference:**
 - File Handling in C - Tutorialspoint

9. Dynamic Memory Allocation

- **Topics:**
 - Dynamic memory allocation using malloc(), calloc(), realloc()
- **YouTube Tutorial:**
 - [Dynamic Memory Allocation in C by GeeksforGeeks](#)
- **Reference:**
 - Dynamic Memory Allocation in C - GeeksforGeeks



System Programming

Goal: Learn system-level programming techniques, including working with the operating system, handling processes, and working with system calls.

10. Introduction to System Programming

- **Topics:**
 - What is system programming?
 - System calls and APIs
- **YouTube Tutorial:**
 - [System Programming Basics by MyCodeSchool](#)
- **Reference:**
 - Introduction to System Programming - GeeksforGeeks

11. Process Management

- **Topics:**
 - Processes and threads
 - Process synchronization and communication
- **YouTube Tutorial:**
 - [Processes and Threads in C by GeeksforGeeks](#)
- **Reference:**
 - Process Management in C - GeeksforGeeks

12. Inter-Process Communication (IPC)

- **Topics:**
 - Pipes, message queues, and shared memory
- **YouTube Tutorial:**
 - [IPC in C by TutorialPoint](#)
- **Reference:**
 - Inter-Process Communication - GeeksforGeeks

13. System Calls and Operating System Interface

- **Topics:**
 - Understanding system calls and interacting with the OS
- **YouTube Tutorial:**
 - [System Calls in C by LearnCodeOnline](#)
- **Reference:**
 - System Calls in C - Tutorialspoint



Recommended Books for C Mastery

- *"The C Programming Language"* – Brian W. Kernighan, Dennis M. Ritchie
- *"C Programming: A Modern Approach"* – K. N. King
- *"Advanced Programming in the UNIX Environment"* – W. Richard Stevens



Best Online Resources

- [C Programming Language](#) - GeeksforGeeks
- [System Programming](#) - Coursera

Note: To master C and system programming, work on building real-world system-level programs, such as a shell or memory manager. The more you practice, the more proficient you'll become at understanding how the operating system works and interacting with it.